A parabola is symmetric with respect to a line called the axis of symmetry. The point of intersection of the parabola and the axis of symmetry is called the vertex of the parabola. The graph of any second degree polynomial gives a curve called "parabola".

Hint : For a quadratic equation , the axis is given by $x = \frac{-b}{2a}$ and the vertex is given by $\left(\frac{-b}{2a}, \frac{-\Delta}{4a}\right)$ where $\Delta = b^2 - 4ac$ is the discriminant of the quadratic equation $ax^2 + bx + c = 0$. Where a $\neq$ 0.

We have already studied how to find the roots of any quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ and $a \neq 0$ theoretically. In this section, we will learn how to solve a quadratic equation and obtain its roots graphically.

3.8.1 Finding the Nature of Solution of Quadratic Equations Graphically

To obtain the roots of the quadratic equation $ax^2 + bx + c = 0$ graphically, we first draw the graph of $y = ax^2 + bx + c$.

The solutions of the quadratic equation are the x coordinates of the points of intersection of the curve with X axis.

To determine the nature of solutions of a quadratic equation, we can use the following procedure.

(i) If the graph of the given quadratic equation intersect the X axis at two distinct points, then the given equation has two real and unequal roots.

(ii) If the graph of the given quadratic equation touch the X axis at only one point, then the given equation has only one root which is same as saying two real and equal roots.

(iii) If the graph of the given equation does not intersect the X axis at any point then the given equation has no real root.

Example 3.51 Discuss the nature of solutions of the following quadratic equations.
(i) $x^2 + x - 12 = 0$ (ii) $x^2 - 8x + 16 = 0$ (iii) $x^2 + 2x + 5 = 0$

Solution

(i) $x^2 + x - 12 = 0$

Step 1: Prepare the table of values for the equation $y = x^2 + x - 12$.

x	−5	−4	−3	−2	−1	0	1	2	3	4
y	8	0	−6	−10	−12	−12	−10	−6	0	8

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